FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation

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Abstract

Federated recommendation (FR) enables collaborative model training while keeping user interactions local. However, transferring recommendation knowledge across services remains challenging due to disjoint item catalogs and uneven target-domain supervision. Frequently observed items provide abundant interaction evidence, whereas low-frequency items offer limited supervision, and client-specific preferences further introduce heterogeneity into semantic adaptation. Existing approaches commonly employ shared transformations or client-level personalization without explicitly accounting for differences in item-level supervision. To address these challenges, we propose FedPASA, a reliability-aware semantic adaptation framework for cross-domain federated recommendation. FedPASA introduces Popularity-Gated Dual-path Routing (PGDR), which combines two residual adaptation paths using item-frequency-derived weights. A private client profile further conditions the adapted representations through feature-wise scaling and shifting, while AutoAnchor incorporates pretrained semantic information through catalog-level anchoring. Extensive experiments on real-world cross-domain recommendation benchmarks demonstrate the effectiveness of FedPASA. The proposed framework achieves superior average cross-domain recommendation performance over competitive baselines under both sampled and full-catalog ranking. Further analyses demonstrate that frequency-aware routing improves recommendation accuracy beyond additional adapter capacity, while client conditioning and semantic anchoring provide complementary adaptation capabilities.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Distributed computing methodologies.*

Keywords

Federated Recommendation; Cross-Domain Recommendation; Semantic Transfer; Long-tail Recommendation

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1 Introduction

Recommender systems learn user–item preferences from interaction histories [12] and support online services. Centralized training, however, raises concerns about sharing these histories. Federated recommendation (FR) keeps interactions local and aggregates model parameters [1, 26, 42]. Recent work develops client co-clustering [13], distributed heterogeneous-graph learning [36], and communication-efficient low-rank updates [28], advancing collaborative recommendation without pooling raw histories [9].

Cross-domain transfer is another longstanding need: platforms wish to reuse a mature catalog to bootstrap a new service with limited behavioral evidence. Many cross-domain methods transfer preferences through overlapping users or aligned representations [3, 25, 37]. Such links may be unavailable across services, while federated constraints prevent pooling user histories. We study disjoint item catalogs without requiring linked users between source and target domains.

Text-based transferable representations offer an alternative to identifier alignment [14, 16, 41]. However, semantic representations alone do not resolve uneven target supervision. Interaction imbalance and conformity can distort representation learning [37, 38]: frequent items supply many ranking constraints, whereas rare items provide little adaptation evidence. Clients also observe different semantic regions [34], so a globally rare item may be familiar to one client yet unseen by another. Excessive adaptation can move sparse items away from useful pretrained semantics. Adaptation should therefore account for item-level evidence, private client context, and the amount of semantic prior to preserve.

Hierarchical personalization, personalized adapters, and expert mixtures address client variation [21, 35, 39]. Client-level personalization [22, 40] alone does not specify how items with different training frequencies should share adaptation paths: a shared transformation uses the same parameterization for all items, while adding an adapter increases capacity without determining its allocation. We instead use item frequency as an observable proxy for target supervision, route items between two residual paths, and condition the routed representation on the local interaction profile.

To address the challenges mentioned above, we propose FedPASA, which combines three operations in a shared semantic adapter (Figure 1). Specifically, Popularity-Gated Dual-path Routing (PGDR) mixes a source-initialized path and a target-initialized path using a frequency-derived weight; client conditioning produces feature-wise scales and shifts from a locally pooled semantic profile; in addition, AutoAnchor interpolates the resulting representation with the frozen text embedding, with the anchor strength set by a catalog-level user-to-item ratio. Both the source-initialized and target-initialized residual paths are jointly optimized using target-domain interactions.

We evaluate this design on multiple real-world datasets, with controls that separate frequency assignment, model capacity, and final personalization. Our contributions are summarized as follows:

- We propose FedPASA, a reliability-aware semantic adaptation framework that adjusts item representations according to evidence strength at the item, client, and catalog levels.

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The framework addresses uneven target supervision: frequent and sparse items contribute different amounts of evidence, while local interaction histories determine the semantic context in which that evidence is observed. The proposed framework applies to different backbone encoders.

- We design PGDR to combine two trainable residual adaptation paths using frequency-derived weights, without an additional learned routing network. Client conditioning adjusts the combined representations using local semantic profiles, while AutoAnchor controls their mixture with the frozen semantic prior through a catalog-level rule.
- We conduct extensive experiments on multiple real-world datasets. With the same text encoder, FedPASA achieves higher average cross-domain recommendation accuracy than the evaluated baselines. Comparisons that retain the same network structure and parameter count further show that frequency-derived mixing weights outperform using a common pair of weights for all items or shuffling the item–weight assignments. Ablation studies further characterize the effects of individual components.

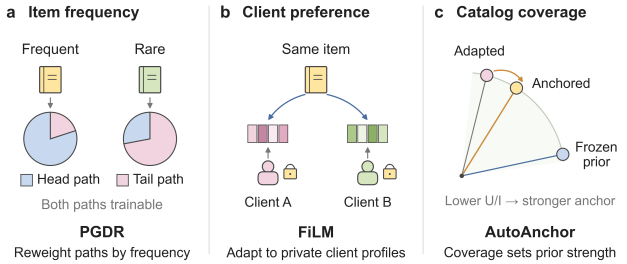


Figure 1: Three adaptation controls: (a) frequency-based path mixing; (b) private-profile conditioning; (c) catalog-dependent prior preservation.

2 Related Work

2.1 Federated Semantic Transfer

Federated recommendation (FR) enables collaborative learning from decentralized user interactions while preserving data privacy [1, 26]. Existing studies address client heterogeneity through proximal optimization and personalized local models [2, 5, 17, 18]. Cross-domain FR further explores knowledge transfer across heterogeneous item catalogs. TransFR combines pretrained language representations, adapter tuning, and local personalization [41]. FedDCSR disentangles shared and domain-specific sequential factors [43], while PFCR and FFMSR exploit textual content for semantic transfer [7, 23]. FedCRF enhances cross-domain knowledge fusion through semantic graphs and contrastive learning [8]. FeDecider further introduces personalized aggregation of directional LoRA updates to address domain heterogeneity [11].

Despite these advances, existing methods primarily focus on semantic representation learning, knowledge alignment, and personalized aggregation, while item-level supervision differences remain less explored. Frequent and infrequent items provide different amounts of interaction evidence, making uniform semantic adaptation insufficient for capturing their distinct learning requirements. To address this limitation, FedPASA introduces popularity-gated dual-path routing (PGDR), which allocates source-initialized and target-initialized adaptation paths according to item frequency.

2.2 Personalization and Preservation

Personalized FR addresses client heterogeneity through adaptation and knowledge sharing. AdapFedRec combines cluster-based negative sampling, weighted client aggregation, and adaptive local differential privacy [10]. FedPA combines private and group adapters with adaptive gating [39], while FedVLR employs personalized multimodal expert fusion [21]. Other studies explore long-tail recommendation through ESAM [4], user preference adaptation through MeLU [15], and expert specialization through multi-gate mixture-of-experts architectures [24]. These approaches generally treat personalization and item imbalance separately. FedPASA coordinates frequency-dependent PGDR with feature-wise modulation from private client profiles based on FiLM [30].

Preserving transferred semantic knowledge is another important aspect of cross-domain adaptation. L2-SP and DELTA retain source information through parameter and feature regularization [19, 20]. However, these methods constrain parameter updates or intermediate representations without directly controlling the contribution of frozen semantic features to final ranking representations. To address this distinction, FedPASA introduces AutoAnchor, which adjusts the semantic prior contribution through catalog-dependent interpolation according to target-domain coverage.

3 FedPASA

3.1 Problem Setup and Overview

We consider cross-domain federated recommendation with disjoint item catalogs. Let C, $\mathcal{U}$, and $\mathcal{I}$ denote the sets of clients, users, and target items, respectively. Each client c maintains a subset of users $\mathcal{U}_c$ and their interaction histories locally. A frozen text encoder maps each item i to a unit-normalized semantic representation $x_i \in \mathbb{R}^d$, providing a common representation space across domains. Each user u is associated with a private embedding $e_u \in \mathbb{R}^d$ that remains on its corresponding client.

FedPASA adapts the frozen semantic representations through three mechanisms: Popularity-Gated Dual-path Routing (PGDR), Private-Profile Client Conditioning (FiLM), and Catalog-Level Semantic Anchoring (AutoAnchor). Source-domain training initializes one residual adaptation path, while the second path is randomly initialized on the target domain. Both paths remain trainable during target-domain federated optimization.

For user u on client c, the recommendation score is defined as

$$s(u,i) = e_u^\top z_{c,i}^{\text{anchor}}. \tag{1}$$

The model is optimized using Bayesian Personalized Ranking (BPR) [32]:

$$\mathcal{L}_{\text{BPR}} = -\mathbb{E}_{(u,i,j)} \log \sigma\big(\tau[s(u,i) - s(u,j)]\big),$$

where i is an observed positive item, j a sampled negative item, and τ controls the score scale.

Figure 2 illustrates the overall FedPASA framework, and Table 5 summarizes the notation. The forward computation consists of

three consecutive stages:

$$x_i \xrightarrow{\text{routing}} h_i \xrightarrow[\text{and residual}]{\text{FiLM}} z_{c,i} \xrightarrow{\text{anchoring}} z_{c,i}^{\text{anchor}}. \tag{2}$$

3.2 Popularity-Gated Dual-path Routing

Target items exhibit substantially different interaction frequencies, resulting in different amounts of supervision for semantic adaptation. PGDR explicitly uses item frequency to determine the relative contributions of two residual adaptation paths, enabling frequency-dependent representation learning within a shared model.

Let $\text{freq}(i)$ denote the number of target-domain training interactions containing item i. We first construct an inverse-frequency score $\bar{q}_i = 1/\sqrt{\text{freq}(i)+1}$ and normalize it over the target catalog:

$$q_i = \frac{\bar{q}_i - \min_j \bar{q}_j}{\max_j \bar{q}_j - \min_j \bar{q}_j + \epsilon}. \tag{3}$$

A threshold t and exponent κ further reshape the routing score:

$$q_i' = \text{clip}\big(\tfrac{q_i - t}{1-t}, 0, 1\big)^{\kappa}.$$

The resulting score q_i' increases as item frequency decreases and controls the relative contribution of the target-initialized path.

PGDR contains two residual adapters, denoted by A_h and A_l. The head adapter A_h is initialized from the source-domain adapter, providing transferred semantic adaptation knowledge. The tail adapter A_l is randomly initialized on the target domain and learns complementary transformations from target interactions. Each adapter contains two sequential residual blocks, $v \leftarrow v + F_\ell(v)$, where F_ℓ is a bottleneck multilayer perceptron, followed by ℓ_2 normalization. Both adapters are optimized during target-domain federated training.

The routed item representation is

$$h_i = (1 - q_i')\, A_h(x_i) + q_i'\, A_l(x_i). \tag{4}$$

Frequently observed items place greater weight on the source-initialized head path, whereas lower-frequency items receive a larger contribution from the target-initialized tail path. The mixed representation h_i is subsequently passed to the client-conditioning module.

3.3 Private-Profile Client Conditioning

Item frequency characterizes supervision at the item level, while clients may observe substantially different semantic regions of the target catalog. To incorporate this client-side heterogeneity, FedPASA constructs a private semantic profile from each client's local interaction history and uses it to condition the routed representation.

For client c, let $\mathcal{I}_c$ denote the multiset of items appearing in its training interactions. Repeated interactions retain their multiplicity in the profile construction. The client profile is defined as

$$p_c = \text{norm}\Big(\tfrac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} x_i\Big). \tag{5}$$

A shared bottleneck network $g_\phi : \mathbb{R}^d \to \mathbb{R}^{2d}$ maps the profile into feature-wise FiLM parameters:

$$(\gamma_c, \beta_c) = g_\phi(p_c). \tag{6}$$

The routed representation h_i is then modulated as

$$\begin{aligned} m_{c,i} &= h_i \odot (1 + \lambda \tanh \gamma_c) + \lambda \beta_c, \\ z_{c,i} &= \text{norm}\big((1-\alpha) A_h(x_i) + \alpha m_{c,i}\big). \end{aligned} \tag{7}$$

The coefficient λ controls the strength of client-dependent modulation. The tanh transformation bounds the multiplicative factor to $[1-\lambda, 1+\lambda]$, while the additive term provides learned client-specific feature shifts. The coefficient $\alpha \in [0,1]$ controls the contribution of the conditioned representation relative to the head-path residual.

For the conditioning module, only the shared parameters ϕ participate in federated communication. The client profile p_c is computed from local training interactions and remains on the client.

3.4 Catalog-Level Semantic Anchoring

After frequency-aware routing and client conditioning, FedPASA further integrates the frozen semantic representation into the final ranking representation. AutoAnchor determines the contribution of this semantic prior according to target-catalog coverage.

We characterize target-catalog coverage using the user-to-item ratio

$$\eta_T = |\mathcal{U}_T| / |\mathcal{I}_T|,$$

where $|\mathcal{U}_T|$ denotes the number of observed target-domain users and $|\mathcal{I}_T|$ denotes the target catalog size. Given coverage thresholds $\tau_l < \tau_h$ and maximum anchor weight ρ, AutoAnchor computes

$$\begin{aligned} \rho_{\text{eff}} &= \rho \cdot \text{clip}\Big(\frac{\tau_h - \eta_T}{\tau_h - \tau_l}, 0, 1\Big), \\ z_{c,i}^{\text{anchor}} &= \text{norm}\big((1 - \rho_{\text{eff}})\, z_{c,i} + \rho_{\text{eff}}\, x_i\big). \end{aligned} \tag{8}$$

The effective weight ρ_{eff} controls the contribution of the frozen semantic prior. When $\eta_T \le \tau_l$, the maximum anchor weight ρ is used. As η_T increases between τ_l and τ_h, the prior contribution decreases linearly. For $\eta_T \ge \tau_h$, the final representation relies directly on the adapted representation. Its weight is determined from target-domain catalog statistics and remains fixed throughout federated training and evaluation.

The default catalog-level gate introduces no additional trainable parameters. We additionally evaluate a per-item learnable alternative,

$$\rho_i = \rho\, \sigma(a_i),$$

where $a \in \mathbb{R}^{|\mathcal{I}|}$ contains item-specific logits included in federated aggregation. Table **??** compares catalog-level and per-item anchoring.

3.5 Federated Training and Local Personalization

FedPASA shares $\theta = \{\theta_h, \theta_l, \phi\}$, comprising the two residual adapters and client-conditioning network. Each client c maintains a private user-embedding table $E_c = \{e_u : u \in \mathcal{U}_c\}$.

At round r, clients initialize θ_c from the broadcast parameters $\theta^{(r)}$, retain E_c from the preceding round, and take K BPR steps updating both. Only θ_c is uploaded. Under full participation, the server applies equal-client FedAvg:

$$\theta^{(r+1)} = \frac{1}{|C|} \sum_{c \in C} \theta_c.$$

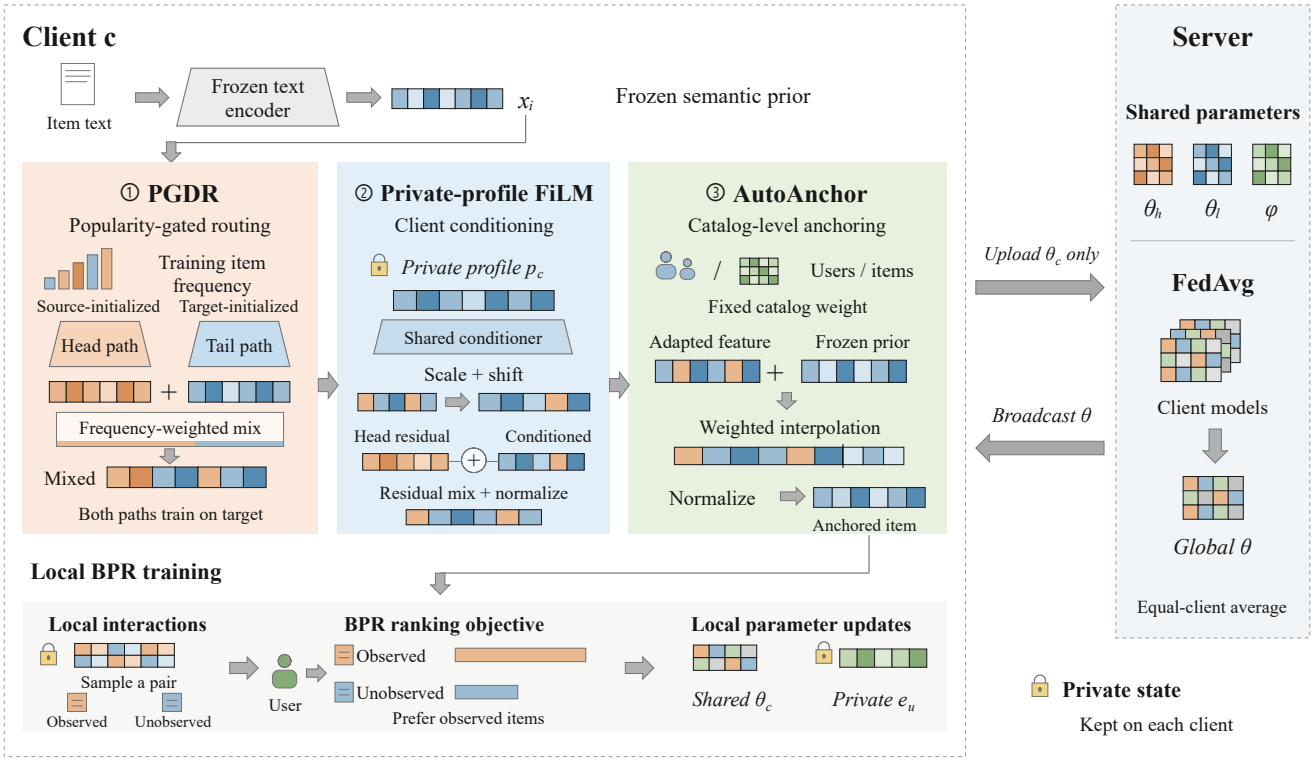


Figure 2: FedPASA framework. PGDR combines source-initialized and target-initialized residual adapters according to item frequency. Private-profile FiLM conditions the resulting representations on local interaction semantics, while AutoAnchor integrates the frozen semantic prior. Clients optimize shared model parameters and private user embeddings using BPR, and the server aggregates the shared parameters through FedAvg.

The federated adaptation stage optimizes the BPR objective without an additional proximal penalty.

After R communication rounds, the resulting global parameters $\theta^{(R)}$ define the shared FedPASA model. We refer to this model before final local personalization as *FedPASA base*. For final personalization, each client initializes its local model from $\theta^{(R)}$, retains its learned private user embeddings, and optimizes

$$\mathcal{L}_{\text{pers},c} = \mathcal{L}_{\text{BPR},c} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2.$$

The proximal term constrains θ_c around the final global solution [17, 18]. Each client takes K_p steps updating both θ_c and E_c; the resulting personalized parameters remain local.

Algorithm 1 summarizes the complete training procedure. Concrete optimization settings are provided in the experimental setup (§4.2).

Initialization and local optimization. Source-domain training uses one residual adapter together with source-domain user embeddings. The resulting adapter parameters initialize the target-domain head path. The tail path is randomly initialized. The final linear layer of the conditioning network is zero-initialized so that client modulation starts from the shared representation.

Target item frequencies are computed exclusively from target training interactions. Each client profile is computed from its own training-item multiset, with repeated interactions retained. Routing statistics, client profiles, and the catalog-level anchor weight are fixed throughout federated optimization, personalization, and evaluation.

Private user embeddings persist across communication rounds. AdamW optimizer states are reset at the beginning of each local optimization phase. During local training, users are sampled uniformly within a client and positive items are sampled uniformly from the corresponding user's training history. Negative items are sampled from items outside the user's observed training interactions. Dropout is enabled during model updates and disabled during evaluation.

The communication process exchanges only the shared adaptation and conditioning parameters. Raw interaction histories, client profiles, and private user embeddings remain on their respective clients.

3.6 Analysis of the Adaptation Controls

We analyze the main adaptation mechanisms from three perspectives: the relationship between target-estimation reliability and semantic-prior weighting, the geometry of normalized semantic anchoring, and the interaction between popularity routing and client conditioning.

Algorithm 1 FedPASA training and local personalization

Require: Frozen item features, training histories, clients C, rounds R, local steps K, personalization steps K_p, proximal coefficient μ_p

For cross-domain training, optimize a source-domain residual adapter by source BPR and equal-client FedAvg
Initialize θ_h from the source adapter; randomly initialize θ_l and ϕ
Compute target routing scores q_i from target training frequencies
Compute each client profile p_c from its training interactions
Compute and fix the catalog-level anchor weight ρ_{eff}
Initialize private user tables $E_c = \{e_u : u \in \mathcal{U}_c\}$ for all clients
$\theta^{(0)} \leftarrow \{\theta_h, \theta_l, \phi\}$
for $r = 0, \ldots, R-1$ **do**
  Broadcast $\theta^{(r)}$ to all participating clients
  for all $c \in C$ **in parallel do**
    $\theta_c \leftarrow \theta^{(r)}$; retain E_c from the preceding round
    Take K local steps on $\mathcal{L}_{\text{BPR}}$, updating (θ_c, E_c)
    Keep E_c local and upload θ_c
  end for
  $\theta^{(r+1)} \leftarrow |C|^{-1} \sum_{c \in C} \theta_c$
end for
Preserve $\theta^{(R)}$ as the FedPASA base model
for all $c \in C$ **do**
  $\theta_c \leftarrow \theta^{(R)}$; retain the final E_c
  $\ell_c \leftarrow \mathcal{L}_{\text{BPR}} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2$
  Take K_p local personalization steps, updating (θ_c, E_c)
end for
return Personalized (θ_c, E_c) stored on each client

3.6.1 *Bias–Variance Tradeoff of Prior Weighting.*

Proposition 1 (Bias–variance tradeoff). *Let z^* denote an ideal target representation and x a semantic prior. Consider an estimated target representation $\hat{z} = z^* + \varepsilon$, where $\mathbb{E}\varepsilon = 0$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma^2/r$ with $r, \sigma^2 > 0$. Here, r characterizes the reliability of the target estimate: larger r corresponds to lower estimation variance. Let $b^2 = \|x - z^*\|_2^2$ denote the squared bias of the semantic prior. For the interpolated estimate $\tilde{z}_\omega = (1-\omega)\hat{z} + \omega x$, we have*

$$\mathbb{E}\|\tilde{z}_\omega - z^*\|_2^2 = (1-\omega)^2\sigma^2/r + \omega^2 b^2, \qquad \omega^* = \frac{\sigma^2}{\sigma^2 + rb^2}. \tag{9}$$

When $b > 0$, ω^* decreases as r increases.

The optimal weight increases with estimation noise and decreases with prior bias: when target evidence is limited, more of a useful prior should be retained. AutoAnchor operationalizes this principle at the catalog level by using target-catalog coverage to determine the contribution of the frozen semantic prior.

3.6.2 *Geometry of Normalized Semantic Anchoring.*

Proposition 2 (Normalized anchor geometry). *For fixed unit vectors x and z, define $w_\omega = (1-\omega)z + \omega x$, $y_\omega = w_\omega/\|w_\omega\|_2$, and $a = x^\top z$. For $0 \le \omega < 1/2$,*

$$\|w_\omega\|_2 \ge 1 - 2\omega > 0, \qquad \frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{\|w_\omega\|_2^3} \ge 0. \tag{10}$$

Therefore, $1 - x^\top y_\omega$ is non-increasing with respect to the anchor weight ω.

This proposition characterizes the geometric effect of normalized semantic interpolation: for a fixed adapted representation, the cosine distance to the frozen prior is non-increasing with the anchor weight. The experimental analysis (§4.4.2) further examines the representation geometry produced after federated optimization.

Table 1: Statistics of the datasets used in evaluation.

Dataset	# Users	# Items	# Interactions
Music	12,315	10,680	339,923
Books	262,698	170,389	10,848,388
Movies	28,226	19,252	1,056,608

3.6.3 *Client Conditioning.*

The multiplicative part of the FiLM transformation in Eq. (7) is $1 + \lambda \tanh \gamma_c$. Since $\tanh \gamma_c \in [-1, 1]$, the resulting feature-wise multiplicative factors satisfy $1 - \lambda \le 1 + \lambda \tanh \gamma_c \le 1 + \lambda$. Thus, λ directly controls the range of client-dependent multiplicative modulation, while the learned additive term $\lambda\beta_c$ shifts the routed representation according to the client's semantic profile.

3.6.4 *Interaction of Routing and Client Conditioning.*

PGDR and client conditioning act on the same semantic representation but control different aspects of adaptation. To make the interaction explicit, consider the case in which FiLM modulation is disabled, i.e., $\lambda = 0$. Combining Eqs. (4) and (7) gives

$$z_{c,i} = \text{norm}\big((1 - \alpha q_i')A_h(x_i) + \alpha q_i' A_l(x_i)\big). \tag{11}$$

The effective coefficient of the target-initialized path is therefore $\alpha q_i'$. The item-frequency-derived score q_i' determines how the two adaptation paths are allocated to each item, while α controls the overall strength of this allocation. Even for the largest routing score $q_i' = 1$, the head path retains a contribution of $1 - \alpha$.

When FiLM is enabled, client-dependent feature scales and shifts are applied to the frequency-routed representation before normalization and semantic anchoring. Consequently, item frequency controls path combination, whereas the private client profile controls feature-level adjustment.

4 Experiments

We evaluate FedPASA through four research questions:

- **RQ1:** How does FedPASA compare with federated recommendation and semantic-transfer baselines?
- **RQ2:** How much does each component contribute beyond additional adapter capacity?
- **RQ3:** How do key hyperparameters affect the recommendation performance of FedPASA?
- **RQ4:** How does FedPASA retain private client information during collaborative learning?

4.1 Datasets

We evaluate cross-domain recommendation on three Amazon datasets. The benchmark contains all six ordered transfers among Amazon Music (CDs & Vinyl), Books, and Movies (Movies & TV) [29]; their item catalogs are disjoint and no linked users are required. Table 1 summarizes the retained data.

4.2 Experimental Settings

4.2.1 *Implementation Details.*

We treat the retained review interactions as implicit feedback and exclude users with fewer than three interactions in each domain. Interaction histories are sorted chronologically, with the latest interaction held out for testing, the

Table 2: Overall performance comparison on three cross-domain transfer tasks. The best and second-best results are highlighted in bold and underline, respectively.

Transfer	Music→Books		Music→Movies		Books→Movies	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [11]	0.2011	0.0714	<u>0.4278</u>	<u>0.1823</u>	<u>0.3981</u>	<u>0.1705</u>
FedMF [1]	0.2005	0.0715	0.2127	0.0792	0.2127	0.0792
FedAvg [26]	0.1949	0.0672	0.3440	0.1388	0.3440	0.1388
FedProx [18]	0.1998	0.0705	0.3381	0.1351	0.3381	0.1351
FedAdam [31]	0.1986	0.0690	0.3398	0.1359	0.3398	0.1359
TransFR [41]	<u>0.2058</u>	<u>0.0723</u>	0.3292	0.1283	0.3330	0.1307
L2-SP [19]	0.2032	0.0718	0.3244	0.1244	0.3141	0.1229
FedPer [2]	0.2031	0.0714	0.3410	0.1353	0.3492	0.1397
pFedMe [5]	0.2047	0.0719	0.3284	0.1285	0.3368	0.1332
FedMoE [27]	0.2039	0.0707	0.3424	0.1355	0.3566	0.1444
FedPASA	**0.2719**	**0.1159**	**0.4980**	**0.2254**	**0.4948**	**0.2284**

Table 3: Overall performance comparison on the remaining three cross-domain transfer tasks and the average results across all six transfer directions.

Transfer	Books→Music		Movies→Books		Movies→Music		Six-transfer mean	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [11]	<u>0.5355</u>	<u>0.2283</u>	0.2058	<u>0.0734</u>	<u>0.5380</u>	<u>0.2284</u>	<u>0.3844</u>	<u>0.1590</u>
FedMF [1]	0.2212	0.0826	0.2005	0.0715	0.2212	0.0826	0.2115	0.0778
FedAvg [26]	0.5074	0.2129	0.1949	0.0672	0.5074	0.2129	0.3488	0.1396
FedProx [18]	0.5024	0.2093	0.1998	0.0705	0.5024	0.2093	0.3468	0.1383
FedAdam [31]	0.5073	0.2098	0.1986	0.0690	0.5073	0.2098	0.3485	0.1382
TransFR [41]	0.5046	0.2121	0.2001	0.0705	0.5048	0.2115	0.3462	0.1376
L2-SP [19]	0.4996	0.2083	0.2017	0.0710	0.5040	0.2122	0.3412	0.1351
FedPer [2]	0.5052	0.2150	0.2040	0.0721	0.5027	0.2099	0.3509	0.1406
pFedMe [5]	0.5054	0.2131	0.2026	0.0712	0.5064	0.2117	0.3474	0.1383
FedMoE [27]	0.5088	0.2143	<u>0.2064</u>	0.0731	0.5107	0.2141	0.3548	0.1420
FedPASA	**0.5408**	**0.2403**	**0.2824**	**0.1249**	**0.5462**	**0.2439**	**0.4390**	**0.1965**

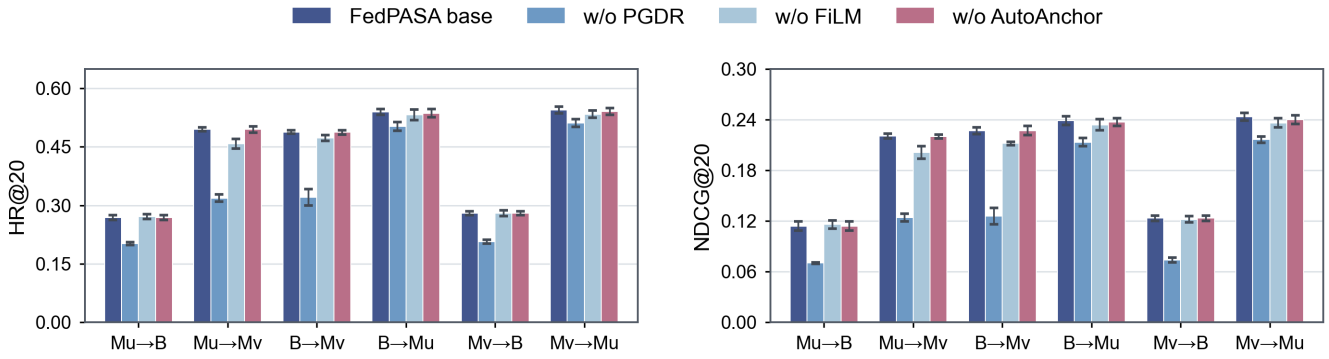


Figure 3: Ablation study across six cross-domain transfer tasks. Error bars indicate standard deviations over five runs. Mu, B, and Mv denote Music, Books, and Movies, respectively.

preceding interaction for validation, and the remaining interactions used for training. We evaluate HR@20 and NDCG@20 under both sampled and full-catalog ranking. Sampled ranking pairs each test item with 99 randomly sampled negative items, excluding the user's training and validation items.

4.2.2 Baselines. We compare FedPASA with representative recent and classical federated recommendation baselines. **FeDecider** [11]

Table 4: Frequency–exposure ablation under full-catalog ranking before personalization. C denotes low-frequency items (≤ 5) observed locally, U denotes low-frequency items (≤ 5) unobserved locally, and W denotes higher-frequency items (> 5).

Model	C: low-frequency / seen		U: low-frequency / unseen		W: frequency > 5	
	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FedPASA	**0.0160**	**0.0070**	**0.0347**	**0.0151**	**0.0102**	**0.0037**
w/o PGDR	0.0041	0.0010	0.0039	0.0013	0.0023	0.0007
w/o FiLM	0.0041	0.0018	0.0276	0.0123	0.0037	0.0014
w/o AutoAnchor	0.0082	0.0023	0.0144	0.0047	0.0039	0.0014

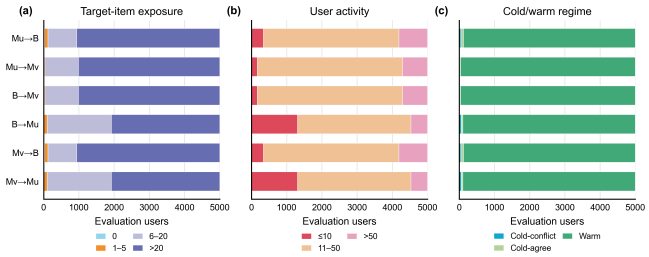


Figure 4: Evaluation-user composition by (a) target-item frequency, (b) user activity, and (c) cold/warm groups across six transfer directions. Mu, B, and Mv denote Music, Books, and Movies, respectively.

employs LLM representations and personalized directional LoRA aggregation for cross-domain recommendation. **TransFR** [41] transfers pretrained language-model adapters and performs local personalization. **FFMSR** [23] leverages textual semantics for federated cross-domain recommendation. **FedDCSR** [43] disentangles shared and domain-specific representations for sequential recommendation. **FedMoE** [27] employs heterogeneous mixture-of-experts models with personalized expert selection and modular aggregation. **FedAdam** [31] applies adaptive server-side optimization, while **FedProx** [18] introduces proximal regularization to address client heterogeneity. **pFedMe** [5] learns personalized federated models through Moreau-envelope optimization. **FedPer** [2] combines globally shared representations with client-specific personalization layers. **L2-SP** [19] regularizes target parameters toward their source initialization. **FedAvg** [26] aggregates local model updates to learn a shared global model, while **FedMF** [1] performs federated matrix factorization for collaborative recommendation.

4.3 Main Results and Discussion (RQ1)

Evidence-dependent adaptation improves transfer across disjoint catalogs. FedPASA achieves the highest HR and NDCG on all six sampled-ranking transfers (Tables 2 and 3), improving mean HR by 14.2% over FeDecider. The largest gains occur on Books and Movies. PGDR allocates adaptation paths according to the available item evidence, while private-profile conditioning incorporates each client's semantic context. The gains in HR and NDCG indicate improved retrieval of relevant items and their placement near the top of the ranked list.

The advantage extends beyond sampled candidates. FedPASA also achieves the highest HR and NDCG in the full-catalog comparison (Tables 6–7). Here, each held-out item competes with the full set of eligible catalog items. The consistent improvements under sampled and full-catalog evaluation demonstrate that FedPASA maintains its ranking advantage across different candidate-set sizes. The shared semantic space supports ranking across disjoint catalogs, while target interactions guide evidence-dependent adaptation.

4.4 Ablation Study (RQ2)

Figure 3 evaluates the base model and its component ablations before local personalization, retaining the BPR objective and remaining modules. Tables 2–3 evaluate the final personalized models.

(1) w/o PGDR. Bypassing the tail path makes routing independent of item frequency:

$$q'_i = 0, \qquad h_i = A_h(x_i). \tag{12}$$

Both metrics decrease on every transfer, consistent with losing item-dependent path allocation. PGDR also exceeds constant and shuffled routing (Table 8), which retain both paths but remove item dependence or break the frequency–weight correspondence. These controls link the gain to frequency assignment.

(2) w/o FiLM. Setting $\lambda = 0$ gives $m_{c,i} = h_i$ and

$$z_{c,i} = \mathrm{norm}\big((1 - \alpha q'_i)A_h(x_i) + \alpha q'_i A_l(x_i)\big). \tag{13}$$

Each item now shares one representation across clients. Restoring FiLM improves accuracy on Movies and Music by introducing profile-specific feature scales and shifts. This modulation complements item-level routing with the semantic context of local interactions.

(3) w/o AutoAnchor. Removing the prior mixture yields

$$\rho_{\text{eff}} = 0, \qquad z^{\text{anchor}}_{c,i} = z_{c,i}. \tag{14}$$

This variant ranks items using the adapted representation. The complete model achieves higher six-transfer mean HR and NDCG (Table 8). AutoAnchor combines the learned target features with pretrained semantic information, using catalog coverage to set their relative contributions.

4.4.1 Evidence Mechanism Analysis. Table 4 examines the effects of different adaptation components across item-frequency and local-exposure groups. PGDR improves HR and NDCG across all three groups, highlighting the effectiveness of frequency-dependent path allocation. The results further illustrate how semantic adaptation responds to different levels of interaction evidence and client familiarity.

Evaluation composition motivates group-wise analysis. Figure 4 summarizes the evaluation-user distribution across six transfer directions. Most evaluation users have frequent target items (> 20 training occurrences), moderate activity (11–50 training interactions),

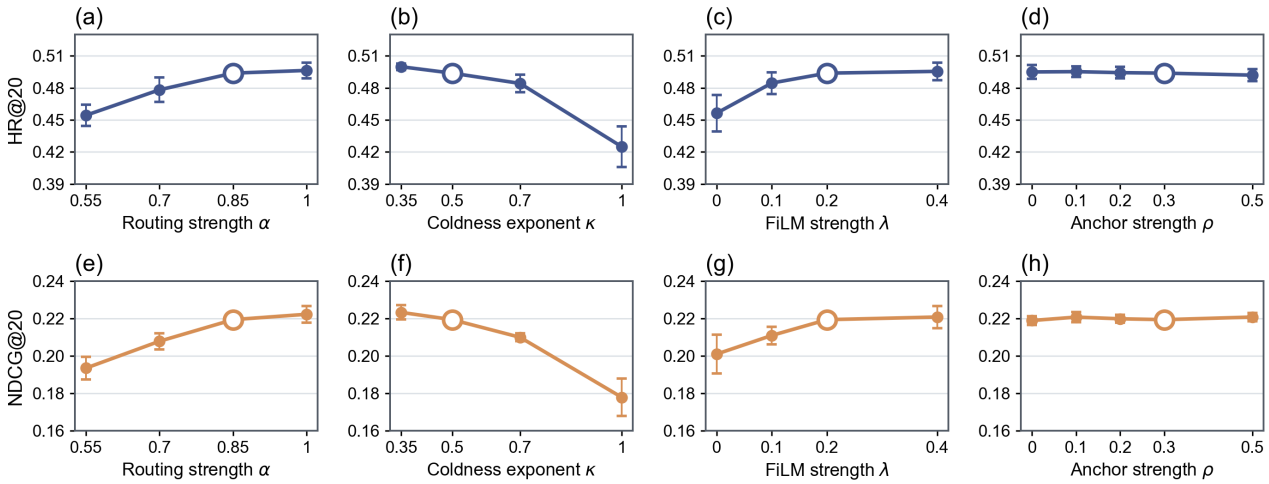


Figure 5: Hyperparameter sensitivity of FedPASA on Music→Movies. The upper and lower panels show HR@20 and NDCG@20, respectively. Error bars indicate standard deviations over three runs, and open circles mark the validation-selected values.

and warm targets (frequency > 5). Transfers to Music contain larger proportions of users with target-item frequencies of 6–20 and low activity (≤ 10 interactions) than transfers to Books or Movies. The predominance of warm targets motivates reporting group-wise performance alongside aggregate metrics (Table 4).

4.4.2 AutoAnchor and representation drift. We measure representation drift through cosine distance from the frozen prior:

$$D = \frac{1}{|C||\mathcal{I}|} \sum_{c \in C} \sum_{i \in \mathcal{I}} \left(1 - \cos(z_{c,i}^{\text{anchor}}, x_i)\right). \tag{15}$$

After training, FedPASA has a smaller cold-item distance and a larger all-item distance than the no-anchor variant. Proposition 2 characterizes the interpolation operation itself: increasing the anchor weight yields non-decreasing alignment with the prior for fixed adapted features.

4.5 Hyperparameter Sensitivity

Figure 5 evaluates routing strength α, coldness exponent κ, client-conditioning strength λ, and anchoring strength ρ on Music→Movies.

Increasing α improves both metrics, with diminishing gains at higher values, supporting greater use of the routed, client-conditioned representation. Larger κ instead reduces accuracy: it suppresses intermediate routing scores and shifts weight toward the source-initialized path, consistent with insufficient target-specific adaptation.

Client conditioning improves performance before leveling off around $\lambda = 0.2$, suggesting that moderate profile-dependent modulation captures most of the benefit. Performance remains stable across the tested anchoring strengths, indicating lower sensitivity to ρ on this transfer. These trends favor prioritizing routing parameters during tuning.

4.6 Privacy Preserving Analysis (RQ4)

In the proposed FedPASA, privacy is supported by separating shared adaptation parameters from private client information. Specifically, after completing local training, each client uploads the parameters of the two residual adapters and the shared FiLM network. The server aggregates these parameters through FedAvg and broadcasts a single global model for the next round. Raw interaction histories, client profiles, and user embeddings remain on their respective clients, enabling collaborative representation learning without pooling users' behavioral records.

For personalized adaptation, each client computes its semantic profile and FiLM modulation locally. Final personalization further updates the model and user embeddings without another upload, retaining client-specific representations on the client.

For stronger privacy protection, FedPASA can be combined with local differential privacy (LDP) mechanisms [6]. Specifically, clients can clip and perturb shared model updates before uploading them to the server, providing additional protection against information leakage during federated optimization.

5 Conclusion

We propose FedPASA for adapting semantic representations across disjoint catalogs under uneven target supervision. PGDR allocates two trainable residual paths by item frequency, private-profile conditioning captures client preferences, and AutoAnchor controls the frozen prior's contribution through catalog coverage. Experiments across six transfer directions show improved average accuracy under sampled and full-catalog ranking. Equal-capacity controls identify frequency-dependent routing as the main source of improvement; component and group analyses characterize the contributions of conditioning and anchoring. Sensitivity analysis further highlights the importance of routing allocation, with diminishing returns from stronger client conditioning and relatively stable performance across the tested anchoring strengths. These findings favor allocating adaptation capacity according to supervision strength. Local profiles complement this allocation by incorporating client preferences without exchanging behavioral histories. Future work will extend this design to evolving catalogs and dynamic client profiles.

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A Experimental Settings

Data processing and evaluation. We use the filtered Amazon Music, Books, and Movies domains described in Section 4.2; the retained dataset sizes are reported in Table 1. The last two interactions of each user form the validation and test sets, and ranking masks the training and validation items. We evaluate HR@20 and NDCG@20 under sampled and full-catalog ranking. For held-out rank r_u, HR@K averages $\mathbf{1}[r_u \le K]$ and NDCG@K averages $\mathbf{1}[r_u \le K]/\log_2(r_u+1)$. For a partition into nonempty groups $\mathcal{G}$, the reported aggregate is

$$M = \sum_{G\in\mathcal{G}} \frac{|G|}{|\mathcal{U}_{\text{eval}}|} M_G, \qquad M \in \{\text{HR@}K, \text{NDCG@}K\}.$$

Thus, each group mean M_G is weighted by its number of evaluated users.

Feature extraction and FedPASA configuration. DistilBERT [33] uses normalized masked-mean features with at most 96 tokens; LLaMA-2-7B features use a fixed signed block projection to 512 dimensions. Unless stated otherwise, FedPASA uses frozen DistilBERT-base-uncased features ($d = 768$), two residual blocks with bottleneck size 64 and dropout 0.1, and AdamW with learning rate 10^{-3} and weight decay 10^{-4}. Each BPR batch contains 32 positive interactions and three negatives per positive, with logit scale 10. Five clients train for 12×60 communication rounds/local steps, except Books-related settings, which use 8×30. We set $(\alpha, t, \kappa, \lambda) = (0.85, 0, 0.5, 0.2)$ and $(\rho, \tau_l, \tau_h) = (0.3, 1, 1.5)$. FedPASA uses zero proximal regularization before its 40-step local personalization stage with $\mu_p = 0.15$.

Baseline Configuration. Adapter-based comparisons use the same frozen DistilBERT representations, client partitions, data splits, and ranking candidates. FedMF [1] uses 64-dimensional factors and 25×80 communication rounds/local steps. FedProx [18] and L2-SP [19] use regularization coefficients of 0.1 and 0.01, respectively. We also include pFedMe [5] and FedMoE [27] as comparison methods.

B Proofs and Mechanism Analysis

We expand the arguments in §3.6. All vector norms are Euclidean unless stated otherwise. The first two proofs concern prior interpolation; the remaining derivations characterize routing and client conditioning.

B.1 Proof of Proposition 1

Write $\delta = x - z^*$, so that $\|\delta\|_2^2 = b^2$, and let $v = \sigma^2/r > 0$. Since $\hat{z} = z^* + \varepsilon$, the interpolation error is

$$\tilde{z}_\omega - z^* = (1-\omega)\varepsilon + \omega\delta.$$

Expanding the squared norm before taking expectations gives

$$\begin{aligned} R(\omega) &= \mathbb{E}\|\tilde{z}_\omega - z^*\|_2^2 \\ &= (1-\omega)^2\mathbb{E}\|\varepsilon\|_2^2 + \omega^2\|\delta\|_2^2 \\ &\quad + 2\omega(1-\omega)\delta^\top\mathbb{E}\varepsilon \\ &= (1-\omega)^2 v + \omega^2 b^2. \end{aligned} \tag{16}$$

The cross term vanishes because δ is fixed and $\mathbb{E}\varepsilon = 0$. The expectation uses the zero mean and total second moment of the noise. Differentiating yields

$$R'(\omega) = 2[(v+b^2)\omega - v], \qquad R''(\omega) = 2(v+b^2) > 0.$$

Strict convexity makes the stationary point the unique minimizer. Since $b^2 \ge 0$, this point lies in $[0,1]$:

$$\omega^* = \frac{v}{v+b^2} = \frac{\sigma^2}{\sigma^2 + rb^2}. \tag{17}$$

Completing the square further gives

$$R(\omega) = \frac{vb^2}{v+b^2} + (v+b^2)(\omega-\omega^*)^2,$$

which quantifies the excess risk of any other weight. When $b = 0$, the prior equals z^* and $\omega^* = 1$. For $b > 0$,

$$\frac{d\omega^*}{dr} = -\frac{\sigma^2 b^2}{(\sigma^2 + rb^2)^2} < 0. \tag{18}$$

Thus stronger target evidence lowers the optimal prior contribution. For $b > 0$, $\omega^* \to 1$ as $r \to 0^+$ and $\omega^* \to 0$ as $r \to \infty$. AutoAnchor implements this inverse relationship using observed catalog coverage as its target-evidence signal; b^2 and σ^2 parameterize the analysis.

B.2 Proof of Proposition 2

For fixed unit vectors x, z, let $a = x^\top z \in [-1,1]$ and $w_\omega = (1-\omega)z + \omega x$. For $0 \le \omega < 1/2$, the reverse triangle inequality implies

$$d_\omega := \|w_\omega\|_2 \ge (1-\omega)\|z\|_2 - \omega\|x\|_2 = 1 - 2\omega > 0.$$

Hence $y_\omega = w_\omega/d_\omega$ is well defined. Its alignment is $x^\top y_\omega = n_\omega/d_\omega$, with

$$\begin{aligned} n_\omega &= a + \omega(1-a), \\ d_\omega^2 &= (1-\omega)^2 + \omega^2 + 2\omega(1-\omega)a. \end{aligned} \tag{19}$$

Using $n'_\omega = 1 - a$ and $d'_\omega = (2\omega-1)(1-a)/d_\omega$, the quotient rule gives

$$\begin{aligned} \frac{d(x^\top y_\omega)}{d\omega} &= \frac{(1-a)[d_\omega^2 - n_\omega(2\omega-1)]}{d_\omega^3} \\ &= \frac{(1-\omega)(1-a)(1+a)}{d_\omega^3} \ge 0. \end{aligned} \tag{20}$$

The last step uses $d_\omega^2 - n_\omega(2\omega-1) = (1-\omega)(1+a)$. Cosine distance is therefore non-increasing, strictly so for $-1 < a < 1$. At $a = 1$ the vectors coincide; at $a = -1$ the normalized mixture remains z on the stated interval. The result applies pointwise to fixed adapted features and can then be averaged over clients and items.

B.3 Routing Gradients at Fixed Capacity

With FiLM disabled, Eq. (11) gives

$$v_i = (1-a_i)A_h(x_i) + a_i A_l(x_i), \qquad a_i = \alpha q'_i.$$

The frequency-derived q'_i is fixed during optimization. Let J_h, J_l be the parameter Jacobians of the normalized adapter outputs, and let $g_i = \nabla_{v_i}\ell$ for a differentiable downstream loss. The chain rule gives

$$\begin{aligned} \nabla_{\theta_h}\ell &= (1-a_i)J_h(x_i)^\top g_i, \\ \nabla_{\theta_l}\ell &= a_i J_l(x_i)^\top g_i. \end{aligned} \tag{21}$$

Subsequent normalization and anchoring are included in g_i. In particular, for $z_i = v_i/\|v_i\|_2$ with $v_i \ne 0$,

$$g_i = \frac{I - z_i z_i^\top}{\|v_i\|_2}\nabla_{z_i}\ell.$$

The projector removes the component parallel to z_i. Submultiplicativity then implies

$$\|\nabla_{\theta_l}\ell\|_2 \le a_i\|J_l(x_i)\|_{\text{op}}\|g_i\|_2,$$

with the head-path bound using $1 - a_i$. Routing scales each item's gradient contributions at fixed parameter dimensions. Constant routing assigns the same weights to all items; shuffled routing preserves the gate distribution and permutes its association with frequency. The adapter Jacobians and downstream loss gradient determine the remaining factors in each contribution.

B.4 Bounds for Client Conditioning

For a fixed client, write $s_c = \tanh\gamma_c$ and $m_c(h) = h + \lambda(s_c \odot h + \beta_c)$. Since h is a convex combination of unit adapter outputs, $\|h\|_2 \le 1$. Together with $\|s_c\|_\infty \le 1$, this yields

$$\begin{aligned} \|m_c(h) - h\|_2 &\le \lambda\big(\|s_c \odot h\|_2 + \|\beta_c\|_2\big) \\ &\le \lambda(1 + \|\beta_c\|_2). \end{aligned} \tag{22}$$

For two inputs with the same client parameters, the additive shift cancels. With $D_c = \text{diag}(1 + \lambda s_c)$,

$$m_c(h) - m_c(\tilde{h}) = D_c(h - \tilde{h}).$$

Each diagonal entry has absolute value at most $1 + \lambda$, so

$$\|m_c(h) - m_c(\tilde{h})\|_2 \le (1+\lambda)\|h - \tilde{h}\|_2. \tag{23}$$

For $0 \le \lambda < 1$, all entries are also at least $1 - \lambda$, giving the lower bound $(1-\lambda)\|h - \tilde{h}\|_2$. Enabling conditioning changes the residual mixture before normalization by $\alpha[m_c(h) - h]$. Its perturbation is therefore bounded by $\alpha\lambda(1 + \|\beta_c\|_2)$, explicitly relating the two strengths to pre-normalization modulation.

The effect after normalization admits a sharper bound through an exact identity. For residual mixtures $v, \tilde{v}$, write $r = \|v\|_2$, $s = \|\tilde{v}\|_2$, $z = v/r$, and $\tilde{z} = \tilde{v}/s$. If $\min(r,s) \ge \eta > 0$, expansion gives

$$\begin{aligned} \|v - \tilde{v}\|_2^2 &= (r-s)^2 + rs\|z - \tilde{z}\|_2^2 \\ &\ge \eta^2\|z - \tilde{z}\|_2^2. \end{aligned} \tag{24}$$

The nonnegative radial term can be discarded, yielding $\|z - \tilde{z}\|_2 \le \|v - \tilde{v}\|_2/\eta$. Applying this to the conditioned and unconditioned residual mixtures gives

$$\|z_{\text{cond}} - z_{\text{uncond}}\|_2 \le \frac{\alpha\lambda}{\eta}(1 + \|\beta_c\|_2).$$

For any unit reference vector, Cauchy–Schwarz bounds the resulting change in its inner product by the same quantity. This connects the pre-normalization modulation bound to directional changes under the stated norm condition.

C Dataset Statistics and Notation

Table 1 reports the filtered dataset sizes. Figure 8 complements these counts with the training-frequency distributions of the three catalogs and the PGDR weights induced by those frequencies.

Table 5: Notation used in FedPASA.

Symbol	Definition
x_i, e_u	Frozen unit item vector; private user vector
A_h, A_l, F_ℓ	Head/tail adapters; residual bottleneck MLP
q_i, q_i', t, κ	Coldness scores, threshold, and exponent
$h_i, m_{c,i}, z_{c,i}$	Routed, modulated, and normalized features
p_c, g_ϕ	Private profile and shared conditioner
α, λ	Mixture and client-modulation strengths
$\eta_T, \rho, \rho_{\text{eff}}, \tau_l, \tau_h$	Coverage, anchor weights, and thresholds
θ, μ, μ_p	Shared parameters and proximal coefficients
r, b^2, σ^2	Reliability, prior bias, and noise scale

D Full Ranking and Run-to-Run Variation

Tables 6–7 consolidate all six transfers and their mean by metric. Full-ranking scores are percentages; M/B/V denote Music/Books/Movies. Bold and underline indicate the best and second-best results. FeDecider uses three runs and the other methods five; FedPASA includes final personalization.

Table 6: Full-ranking HR@20 (%) across six transfers.

Method	M→B	M→V	B→V	B→M	V→B	V→M	Mean
FedMF [1]	0.0280	0.2160	0.2160	0.2440	0.0280	0.2440	0.1627
FedAvg [26]	0.0160	0.3720	0.3720	1.2480	0.0160	1.2480	0.5453
FedProx [18]	0.0080	0.4080	0.4080	1.2960	0.0080	1.2960	0.5707
FedAdam [31]	0.0080	0.3320	0.3320	1.1000	0.0080	1.1000	0.4800
TransFR [41]	0.0120	0.3680	0.3000	1.2560	0.0040	1.3240	0.5440
L2-SP [19]	0.0040	0.2880	0.2800	1.3200	0.0080	1.3240	0.5373
FedPer [2]	0.0120	0.3400	0.4040	1.2680	0.0200	1.1920	0.5393
pFedMe [5]	0.0080	0.2880	0.3360	1.2480	0.0040	1.3160	0.5333
FedMoE [27]	0.0080	0.4080	0.4480	1.3440	0.0200	1.3680	0.5993
FeDecider [11]	0.0133	0.7133	0.5467	1.3600	0.0133	1.4533	0.6833
FedPASA	**0.1680**	**1.2640**	**1.4000**	**1.7680**	**0.1800**	**1.8800**	**1.1100**

Table 7: Full-ranking NDCG@20 (%) across six transfers.

Method	M→B	M→V	B→V	B→M	V→B	V→M	Mean
FedMF [1]	0.0124	0.0893	0.0893	0.0915	0.0124	0.0915	0.0644
FedAvg [26]	0.0047	0.1453	0.1453	0.4325	0.0047	0.4325	0.1942
FedProx [18]	0.0039	0.1510	0.1510	0.4747	0.0039	0.4747	0.2099
FedAdam [31]	0.0027	0.1103	0.1103	0.4134	0.0027	0.4134	0.1755
TransFR [41]	0.0035	0.1245	0.1074	0.4668	0.0012	0.5002	0.2006
L2-SP [19]	0.0009	0.1083	0.1092	0.4640	0.0019	0.4851	0.1949
FedPer [2]	0.0043	0.1160	0.1388	0.4537	0.0065	0.4216	0.1902
pFedMe [5]	0.0024	0.0937	0.1202	0.4589	0.0011	0.4856	0.1937
FedMoE [27]	0.0021	0.1472	0.1606	0.5012	0.0058	0.5192	0.2227
FeDecider [11]	0.0038	0.2509	0.1921	0.4878	0.0082	0.5490	0.2486
FedPASA	**0.0609**	**0.4552**	**0.5015**	**0.6537**	**0.0869**	**0.6898**	**0.4080**

Figure 9 visualizes the performance differences across target catalogs and the variation across runs underlying Tables 6–7.

Run-to-Run Variation. For sampled-ranking evaluation, the standard deviations of the six-transfer mean HR@20 and NDCG@20 are 0.0058 and 0.0035 for FedPASA, and 0.0070 and 0.0042 for FeDecider, respectively. These values are reported on the original metric scale. Each run is averaged across transfers before computing the standard deviations.

E Additional Ablation Analysis

Table 8 evaluates component removals and alternative routing strategies across six transfer directions. Removing PGDR produces the largest performance reduction relative to the unpersonalized FedPASA base (HR@20 of 0.4360 and NDCG@20 of 0.1944). Removing FiLM and AutoAnchor also reduces both metrics, indicating their complementary contributions to semantic adaptation.

Constant and shuffled routing preserve the dual-adapter architecture while modifying the frequency-dependent assignment. Both alternatives yield substantially lower accuracy, demonstrating the importance of popularity-aware path allocation beyond additional adapter capacity.

Table 8: Component and routing controls across six transfer directions. Results are averaged over five runs. Ablations use the base model; the final FedPASA row includes local personalization.

Model	HR@20	NDCG@20
w/o PGDR	0.3434	0.1373
w/o FiLM	0.4245	0.1867
w/o AutoAnchor	0.4346	0.1934
Constant routing	0.3481	0.1404
Shuffled routing	0.3381	0.1350
FedPASA	**0.4390**	**0.1965**

Matched-Capacity Routing and Local Refinement. Relative to the unpersonalized base, constant and shuffled routing reduce HR@20 by 8.79 and 9.79 percentage points, respectively, while NDCG@20 decreases by 5.40 and 5.94 points. These comparisons highlight the contribution of frequency-dependent routing under fixed adapter capacity. Final local personalization further improves recommendation performance, yielding the highest six-transfer mean HR@20 and NDCG@20 among the evaluated configurations.

F Hyperparameter Selection and Non-IID Robustness

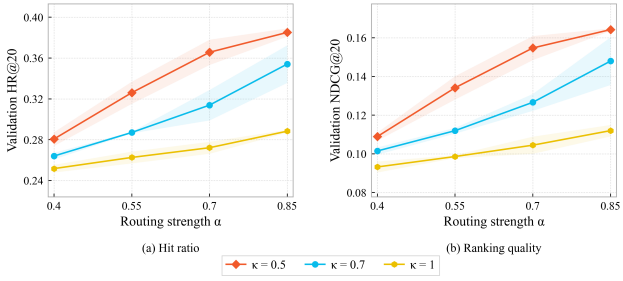


Figure 6: Joint routing validation on Music→Movies: three-run means ± SD; eight rounds, 30 local steps, $\lambda = 0.4$, and $\rho = 0.2$.

The validation search proceeds in two stages. Figure 6 compares routing strength and coldness exponent at fixed $(\lambda, \rho) = (0.4, 0.2)$, selecting $(\alpha, \kappa) = (0.85, 0.5)$. The second stage holds these routing parameters fixed and jointly varies FiLM and anchoring strengths. Both stages use three runs, eight communication rounds, and 30 local steps per round. The selected $(\lambda, \rho) = (0.2, 0.3)$ yields mean validation HR@20 of 0.3925, giving the final configuration $(\alpha, \kappa, \lambda, \rho) = (0.85, 0.5, 0.2, 0.3)$.

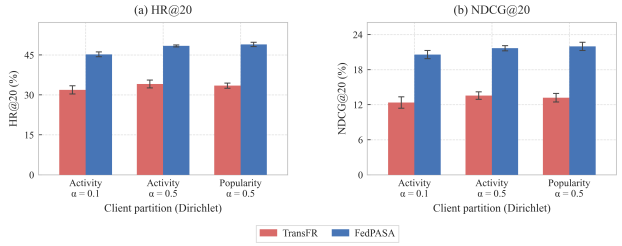


Figure 7: Non-IID Music→Movies: base-model means ± SD (%), three runs, 12×60 rounds/steps. Here α is Dirichlet concentration.

Client Heterogeneity Analysis. Figure 7 evaluates FedPASA under activity-based and popularity-based client partitions with different Dirichlet concentrations. FedPASA consistently outperforms TransFR in both HR@20 and NDCG@20 across the evaluated settings. These improvements demonstrate the effectiveness of the proposed framework under heterogeneous client interaction distributions.

For client updates Δ_c, convexity gives $\|\frac{1}{C}\sum_{c=1}^{C}\Delta_c\|_2^2 \le \frac{1}{C}\sum_{c=1}^{C}\|\Delta_c\|_2^2$, where C is the number of participating clients. Thus equal-client aggregation bounds the global update norm by the root-mean-square client update norm.

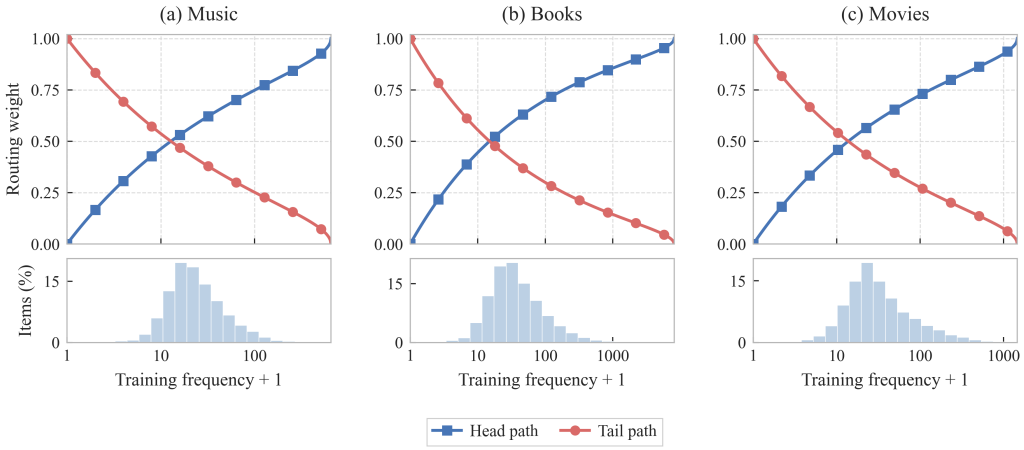


Figure 8: PGDR on three catalogs: head weight $1 - q'_i$ and tail weight q'_i (top); training-frequency distributions (bottom).

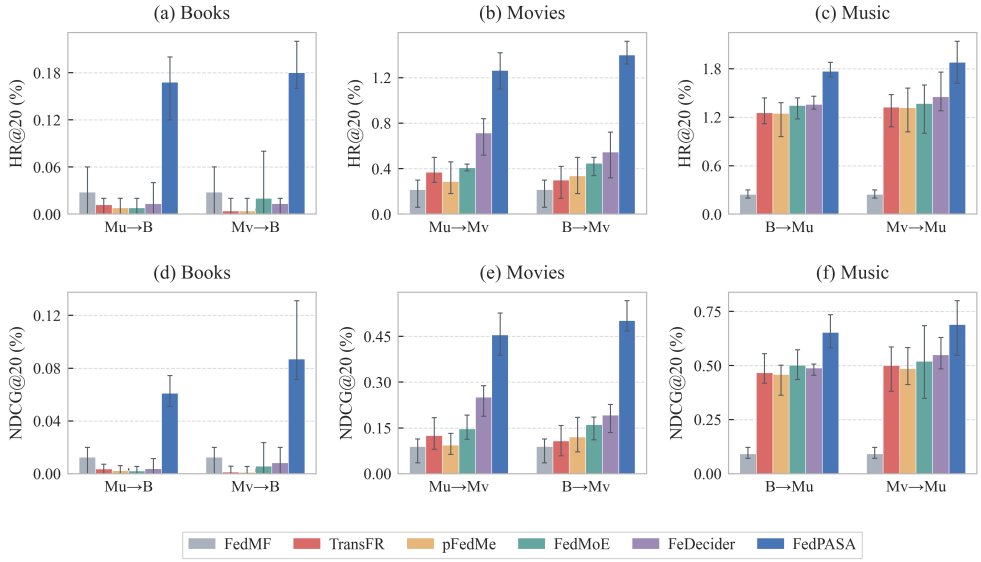


Figure 9: Full-ranking performance across six transfer directions, grouped by target domain. Error bars indicate the minimum and maximum values across runs.

Proof of frequency-monotone routing. For Figure 8, fix the catalog normalization constants and let $D = \max_j \bar{q}_j - \min_j \bar{q}_j + \epsilon > 0$. Assume $0 \le t < 1$ and $\kappa > 0$. For frequency f, set $u(f) = (q(f) - t)/(1 - t)$. In the unsaturated region $0 < u(f) < 1$, differentiation gives

$$\frac{dq'(f)}{df} = -\frac{\kappa u(f)^{\kappa-1}}{2D(1-t)(f+1)^{3/2}} \le 0.$$

Clipping preserves this ordering outside the interior. Thus lower-frequency items receive at least as much tail-path weight. At $t = 0$, $\partial q'/\partial \kappa = q^{\kappa} \log q < 0$ for $0 < q < 1$.

Proof of the mixture norm bound. With FiLM disabled, let $a = \alpha q'_i$ and $c = A_h(x_i)^{\top} A_l(x_i)$. Unit adapter outputs give

$$\|v_i\|_2^2 = 1 - 2a(1-a)(1-c) \ge (1+c)/2.$$

The inequality follows from $a(1-a) \le 1/4$. Hence $c \ge c_0 > -1$ gives $\|v_i\|_2 \ge \sqrt{(1+c_0)/2} > 0$, supplying an explicit condition for Eq. (24).

Transfer-Wise Gains under Full-Catalog Ranking. Tables 6–7 report the full-ranking results across six transfer directions. FedPASA achieves the highest HR@20 and NDCG@20 on all evaluated transfers, with particularly pronounced improvements for transfers targeting Movies. Figure 9 further illustrates these performance differences across target catalogs. The consistent improvements demonstrate FedPASA's ranking effectiveness under full-catalog evaluation.

Proof of coverage-gate stability. For fixed unit x, z, write $\omega_\eta = \rho_{\text{eff}}(\eta)$ and denote their normalized anchor mixture by y_η. For $0 \le \rho < 1/2$, clipping gives $|\omega_\eta - \omega_{\tilde{\eta}}| \le \rho|\eta - \tilde{\eta}|/(\tau_h - \tau_l)$. Each mixture has norm at least $1 - 2\rho$. Equation (24) and $\|x - z\|_2 \le 2$ imply

$$\|y_\eta - y_{\tilde{\eta}}\|_2 \le \frac{2\rho|\eta - \tilde{\eta}|}{(1-2\rho)(\tau_h - \tau_l)}.$$

For fixed e_u, Cauchy–Schwarz bounds the score change by this quantity times $\|e_u\|_2$.